



Land Suitability for Soybean Production in Bavaria, Germany, using Multi-Criteria GIS Analysis

By

Adebola Adedayo

(Student ID: 20250853)

November 3rd, 2025

Land Suitability for Soybean Production in Bavaria, Germany, using Multi-Criteria GIS Analysis

Adebola Adedayo¹

¹NOVA Information Management School,
Campus de Campolide, 1070-312 Lisboa, Portugal.
20250853@novaims.unl.pt

Abstract

Soybean is considered a high-value crop because it is a source of high-quality protein for human and animal diets. The growing need for food in Germany and other parts of Europe, as well as effective agricultural land management, has encouraged the development of land suitability calculation methods. This study aims to delineate soybean (*Glycine max*) land suitability in Bavaria, Germany, using a GIS-based multicriteria approach. A variety of criteria, including terrain (slope, topographic wetness), climate (temperature, precipitation), soil pH, and land use/land cover, were considered for the land suitability analysis to produce soybeans. Each criterion was assigned a weight after carrying out a pairwise comparison with AHP. The calculated weights were used to generate a weighted overlay surface, which serves as a suitability map based on FAO suitability classes (highly suitable to permanently not suitable). Results indicate that 14.16% of the study area is highly suitable for soybean production, 54.89% is moderately suitable, and 27.81% is marginally suitable. The results also indicated that about 3.11% are currently not suitable for soybean cultivation, and about 0.03% are permanently unsuitable for soybean cultivation. Sentinel-2 NDVI at peak season was used for indirect validation, and about 61% of the areas classified as highly suitable areas matched the validation. The final land suitability map will support the targeted expansion and management of soybean cultivation across Bavaria.

Keywords: AHP, land suitability, soybean, weighted overlay

1. Introduction

Soybean (*Glycine max*. (L.) Merr.) is one of the most significant crops worldwide and is cultivated for its high protein and oil content. As a legume, it is used for a variety of human food products, serves as a primary source for livestock feed and can improve soil fertility (Munene et al., 2017). Soybean is a nitrogen-fixing legume that plays a crucial role in organic crop rotations by enriching the soil with rhizobia bacteria, which fix atmospheric nitrogen (Coulter & Moncada, 2008). Regarding crop production in Europe, soybean remains a relatively new crop (Michels & Wever, 2025). While soybean production has increased noticeably, particularly in northern and eastern Europe, many potential sites remain underutilized; this is also the case for Germany (Böttcher & Zimmer, 2021). Yields in the research and farm context across Germany, especially in southern regions, vary from low to moderately high ranges depending on the cultivar, planting time and local conditions, reflecting the strong influence of site factors and management (Fogelberg & Recknagel, 2017).

In recent years, Germany's soybean production area has grown to 44,800 hectares by 2023, mostly in Bavaria and Baden-Württemberg. This results from improved technologies and farming techniques, such as crop rotations and tillage methods, the use of fertilizers, and the use of early-maturing varieties such as *Merlin* and *Primus*, which are bred particularly for Bavaria and central Europe (Fogelberg & Recknagel, 2017; Michels & Wever, 2025). According to a report by Kantar Business Intelligence (Weinzierl, 2019), Bavaria accounts for nearly 50% of the soybean cultivation area in Germany.

The world's growing need for food and bioenergy, coupled with efficient agricultural land management, has prompted the development of techniques for calculating land suitability to promote sustainable environmental practices and efficient agricultural land management (Radočaj et al., 2020). Land

Suitability Evaluation (LSE) is a scientific method that examines the land's suitability and compatibility for particular uses (Bedawi et al., 2016). The goal of Land Suitability Evaluation is to find the best spatial arrangement for sustainable usage based on requirements, preferences, or predictive criteria (Malczewski, 2004). The use of more precise and quantitative components to evaluate the agronomic potential of the biophysical resources has been the focus of advancements in land suitability evaluation (Rosa & C.A., 2009).

Traditionally, in Germany, soybean land suitability evaluation is done using several semi-quantitative land evaluation methods. The Crop Heat Units (CHU) is a temperature-based index used to estimate whether a region has adequate heat to support soybean growth. Precipitation and Water Balance Analysis is also employed to calculate the total rainfall in the area of interest (Fogelberg & Recknagel, 2017). Soil, rainfall, temperature, topography and other non-biophysical elements are important factors that highly influence the land suitability for soybean calculation (Kumar et al., 2013). Therefore, Land suitability assessment is a complex process, and such complexities lead to the simultaneous use of several decision support tools, such as Geographic Information Systems (GIS) and multi-criteria decision analysis (MCDA), which are inevitable in determining suitable land for crop production (Munene et al., 2017).

Multiple techniques have been designed to ascertain the relative relationship between the criteria for suitability calculation, one of the most prominent being the Analytic Hierarchy Process (AHP) (Ishizaka & Labib, 2009). AHP turns expert judgements into defensible criterion weights by pairwise comparisons and a consistency check, reducing subjectivity (Aragonés-Beltrán et al., 2014). After creating data layers for each of the chosen criteria, each data layer is assigned a relative weight and land suitability maps are the result of a weighted overlay analysis using the weighted data layer as inputs (Perveen et al., 2013)

Land Suitability Evaluation is an effective tool for categorising suitable areas by evaluating the characteristics of the land and comparing them to the specifications of a particular land utilisation type (Food and Agriculture Organization of the United Nations (FAO), 2007). In Bavaria, soybeans compete economically with cereals and rapeseed in many sub-regions. Bavaria is agronomically promising for early soybean but exhibits spatial heterogeneity in heat sums, soils, and water supply (Weinzierl, 2019). Therefore, assessing land suitability for soybean production in Bavaria is important; otherwise, introducing new and enhanced technologies won't have any beneficial effects on soybean productivity. Also, a proper, comprehensive land suitability analysis will motivate farmers to implement better management practices and enable them to cultivate soybeans in areas more suited for their growth (Sadiq et al., 2023).

Despite the paucity of research on soybean suitability evaluation in Germany, studies that use GIS-based multi-criteria analysis to improve soybean production in the Free State of Bavaria, Germany, are significantly lacking. Hence, this study aimed to delineate and evaluate land suitability for soybean production in Bavaria, Germany, using a GIS-based multicriteria approach. The objective of this research is to generate a suitability map that could serve as a decision-making tool for choosing appropriate locations for soybean cultivation in Bavaria. Also, an independent accuracy assessment to validate predicted suitability values was performed using a unique peak NDVI method developed from Sentinel-2 multitemporal images. The findings of this land suitability evaluation could help Bavaria produce more soybeans, thereby contributing to Germany's overall goal to reduce reliance on soybean imports.

2. Methods and Data

The proposed method of GIS-based multicriteria analysis for soybean suitability cultivation in Bavaria and its validation consists of six major steps. These are identification and selection of relevant criteria, data preprocessing, weight determination, reclassification, creation of the final suitability surface, and validation. ESRI's ArcGIS Pro software was used in this research for data processing, calculations, and visualization. Microsoft Excel was also utilised for calculating the final weight with the Analytical Hierarchy Process (AHP). The overall methodology flow followed in this study is shown in Figure 2.

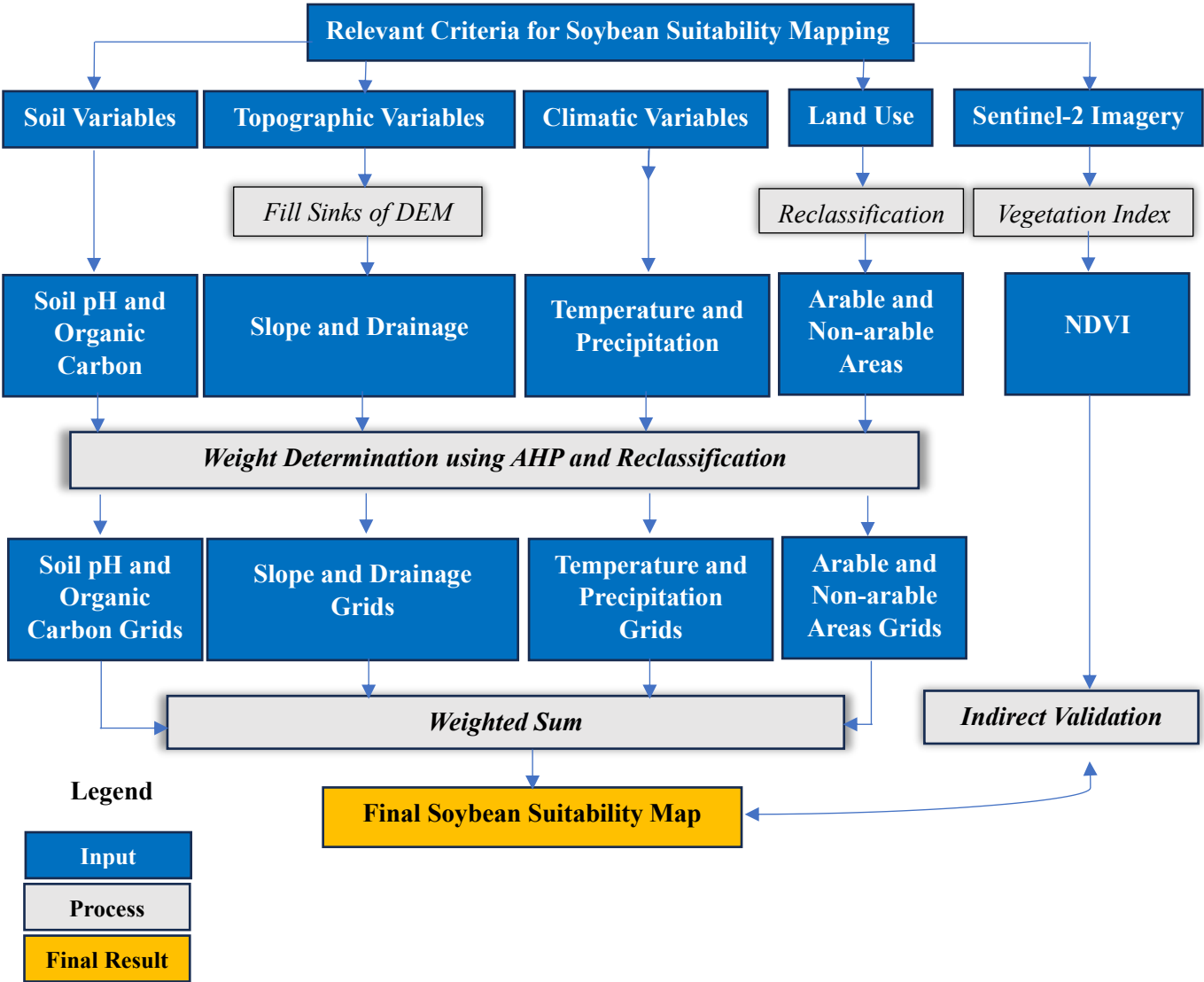


Figure 1. The workflow model showing the methodology of the study.

2.1. Study Area

The Free State of Bavaria is a state in the southeast of Germany, located in Central Europe. It has an area of 70,550km and is considered the largest German State by land area. The study area falls between latitudes 47° 6' - 50° 34' N and longitudes 8° 58' - 13° 51' E (Bogenreuther et al., 2025). It has 71 counties, as shown in Figure 2. In Bavaria, 47 % of its area serves as agricultural land (66 % cropland, 34 % permanent grassland). The predominant soils are Luvisols, Gleysols, Leptosols, Stagnosols, and Cambisols (Zepp et al., 2023).

This region lies in the warm-temperate zone with a mean annual temperature of 8.6 °C (which ranges from 5.3 °C to 10.3 °C) and a mean annual precipitation of 892 mm (which ranges from 583 mm to 1954 mm) (Wiesmeier et al., 2025). The elevation is between 100 m and 1,000 m above sea level (EEA 2023). The mean annual temperature is between 6 °C and 10 °C, and the precipitation is between 551 mm and 1,800 mm per year (DWD 2023).

The State of Bavaria is a net exporter of animal products, and its most commercially significant product is Milk, making up 26.7% of the agricultural gross output in 2018. The production area of legumes, including soybean, lupin, field pea, faba bean, and field bean, has doubled as a result of Bavaria's protein drive, reaching 600 km² in 2019 (Gebhardt et al., 2022).

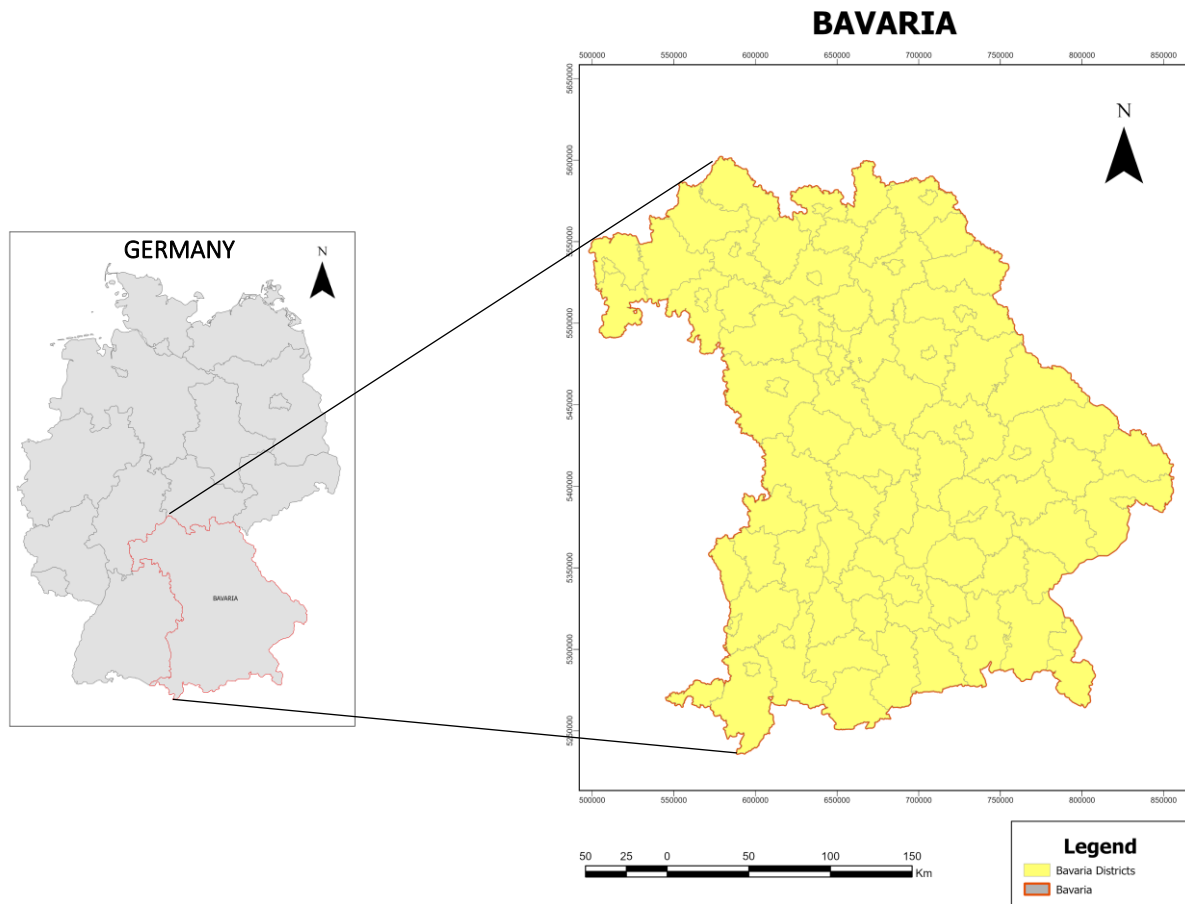


Figure 2. Map of study location.

2.2. Data Sources and Processing

In this study, the selection of the necessary variables for suitability assessment was based on previous research on soybean cultivation. The key factors influencing the suitability of soybean production, such as topography, vegetative characteristics, climate, and land use and land cover (LULC), were obtained as secondary data from credible online sources and are listed in Table 1. All the data layers were initially converted to the ETRS 1989 UTM Zone 32N coordinate system to guarantee consistency and enable additional analysis. In addition, all datasets covering the whole of Europe and Germany were clipped to the study area's boundary, obtained from the Database of Global Administrative Areas (GADM) and

resampled to a resolution of 25 meters to yield a final surface with high resolution and high spatial accuracy (ESRI).

Table 1. Data types and sources.

Parameter	Unit	Data Type	Data source
Soil pH	log	Kriged Raster	European Soil Data Center
Soil Organic Carbon (SOC)	%		
Slope	%	SRTM DEM	European Environment Agency
Drainage	Spatial		
LULC	Spatial	Sentinel-2 Imagery	ArcGIS Living Atlas
Temperature	°C	ESRI Grid	Deutscher Wetterdienst
Precipitation	mm		

2.2.1. Soil data

The soil pH and organic carbon data used in this study were obtained from the European Soil Database, facilitated by the European Soil Data Center as a single reference point for all relevant soil data and information in Europe. The soil pH data were a compilation of 12,333 soil pH measurements across Europe, which were used to produce a raster surface at 1 km resolution based on Regression-Krigging (European Soil Data Centre (ESDAC), 2010).

The soil organic carbon data were generated through the application of digital soil mapping techniques to the first European harmonised geo-referenced topsoil (0–20 cm) database, derived from the Land Use/Cover Area frame statistical Survey (LUCAS) (Brogniez et al., 2015).

2.2.2. Climate Data

The temperature and precipitation data for Germany were obtained from Deutscher Wetterdienst in ESRI Grid format. The climate data used in this study are the Annual sum of monthly precipitation grids in °C and the Annual sum of monthly temperature grids in mm, based on observations made over Germany in the year 2024 at a spatial resolution of 1 km (DWD, 2024).

2.2.3. Remote sensing data and Topographic Data

Sentinel-2 imagery for the year 2024 was obtained from the ArcGIS Living Atlas website. This imagery was used to classify the land use and land cover for the Free State of Bavaria. Four images were downloaded at a resolution of 10 m through ESRI's Land Cover Explorer. On the other hand, elevation data for the study area were extracted from the SRTM Digital Elevation Model (DEM) obtained from the European Environment Agency, as highlighted in Table 1. The DEM was hydrologically corrected by filling all the sinks and further employed to derive the slope and soil drainage (Topographic Wetness Index) of the study area.

Furthermore, a Sentinel-2 imagery obtained from Copernicus and captured in August 2025, during the peak season of soybean growth, was used to compute the Normalized Difference Vegetation Index (NDVI), which was utilized for an indirect validation of the final suitability values. The NDVI is an indicator that makes use of reflectance data from the red and near-infrared (NIR) bands (de la Casa et al., 2018). The NDVI was calculated based on Equation 1.

$$NDVI = \frac{R_{NIR} - R_{RED}}{R_{NIR} + R_{RED}} = \frac{Band5 - Band4}{Band5 + Band4} \quad (1)$$

where NIR is the Near Infrared Band.

2.3. Suitability Assessment

2.3.1. Evaluation of suitability requirements and reclassification of criteria

This study examined soil suitability for soybean cultivation using geostatistical methods in a GIS environment. The soil's suitability was evaluated using the FAO framework for land assessment, which was modified based on improvements recommended by Sys et al. (1991) and includes seven important parameters: soil pH, soil organic carbon (SOC), slope, soil drainage, LULC, temperature and precipitation.

The degree of constraints on the specific requirements listed in Table 2 was taken into consideration when conducting the suitability study for soybeans. As a result, locations in the study area were classified as highly suitable (S1), moderately suitable (S2), marginally suitable (S3), currently not suitable (N1), and permanently not suitable (N2) based on the specifications of FAO (2007).

Table 2. Criteria for evaluating land suitability for soybean in Bavaria, Germany.

Parameters	High Suitability (S1)	Moderate Suitability (S2)	Marginal Suitability (S3)	Current Unsuitability (N1)	Permanent Unsuitability (N2)
pH (log)	6 – 7	5.5 – 6	5.0 – 5.5	4.5 – 5.0	<4.5
SOC (%)	2.4 – 3.5	1.5 – 2.4	0.8 – 1.5	0.4 – 0.8	<0.4
Slope (%)	0 – 2	2 – 4	4 – 6	6 – 8	>8
Drainage (TWI)	0 – 10	10 – 15	15 – 20	20 – 25	>25
LULC	Vegetation	Forest /Rangeland	Bare lands	Flooded Vegetation	Built-up areas/Water
Temperature (°C)	15 – 17	10 – 15	8 – 10	5 – 8	< 5
Precipitation (mm)	600 – 1000	1000 – 1400	597 – 600	1400 – 2000	>2000

Source: Modified from Sys et al. (1991), Sadiq et al. (2023), Munene et al. (2017), and Radočaj et al. (2020).

2.3.2. Factors considered in the suitability analysis

2.3.2.1. Slope

Slope is a crucial topographic factor in land suitability mapping; it serves as a guide for land management and erosion risk management (Munene et al., 2017). It is defined as a measurement that shows the change in steepness or inclination of a given surface over the horizontal plane and can be expressed in degrees or percentage (Wilson & Gallant, 2000). Soil development is closely related to topographical features. Soil thickness increases when the slope angle decreases and decreases when the slope angle increases (Geng et al., 2019). Concurrently, soil erosion occurs in areas with steep slopes, accelerating the loss of soil nutrients. Slope affects several soil characteristics, such as clay material composition, soil moisture content and the availability of other vital elements like calcium, magnesium, and nitrogen (Khormali et al., 2007). Soybeans grow best on land with slopes between 0% and 2% (Sys et al., 1991). In this study, the slope was derived from the DEM using the slope tool on ArcGIS and delineated as a percentage rise.

2.3.2.2. Drainage (Wetness Index)

Drainage is another important topographic factor in land suitability analysis, which provides information about the soil moisture level in an area. Although soybeans may thrive in a variety of soil types, they are extremely vulnerable to waterlogging and cannot survive in areas with continuous water buildup (Sys et al., 1991). The development of crops is hampered by the excessive retention of water in the soil because it blocks airflow, which results in an inadequate supply of oxygen in the roots and the buildup of carbon dioxide (Montgomery et al., 2016). Therefore, improving the drainage system is essential to guaranteeing

adequate aeration in the roots of soybean plants, which promotes ideal plant growth and increases total output.

In this study, soil drainage was represented by the topographic wetness index (TWI), which is a steady-state wetness indicator that reflects soil moisture and drainage conditions. TWI is defined as the ratio of the local upslope contribution area to the slope angle, as determined by the natural algorithm (Wilson & Gallant, 2000). It applies to regions with significant vertical flow and lateral water movement (Schmidt & Persson, 2003). This was the situation in the study area, and it was reasonable to use it to represent drainage. Equation 2 was used to calculate the Topographic Wetness Index (TWI) using the raster calculator on ArcGIS, where A represents the flow accumulation of the area in square meters and B represents the slope in radians. Equation 3 was used to convert the slope from degrees to radians using the raster calculator.

$$TWI = \ln\left(\frac{A}{\tan B}\right) \quad (2)$$

$$\text{Slope in radians} = \text{Slope in degrees} \times \frac{\pi}{180} \quad (3)$$

Before obtaining the flow accumulation (A), the flow direction was initially extracted from the DEM using the flow direction tool. This tool was used to indicate the flow direction of water out of each cell in the DEM. Flow accumulation (A) was then computed by enabling the flow accumulation tool and using the flow direction dataset as the input. The total weight of all cells entering each downslope cell in the output raster is used by the Flow Accumulation tool to compute flow accumulation (Jenson & Domingue, 1988). Some cells that had negative values due to the rate of drainage being greater than the rate of water supply in the wetness index grid were reclassified to zero using the raster calculator, as shown in the expression below

Con (("Old TWI") < 0, 0, ("Old TWI"))

2.3.2.3. Soil pH

The pH level of the soil significantly impacts soil nutrient availability and the overall crop development (Liu et al., 2014). Also, the soil's pH level directly and indirectly affects a variety of physical, chemical, and biological activities in the soil. The ideal soil pH range for soybeans is 6 to 7 (Sadiq et al., 2023).

2.3.2.4. Organic Carbon

The organic carbon (OC) reflects the amount of organic matter present in the soil (Sadiq et al., 2023). OC is an important source of nutrients for crops and meets the nutritional needs of microorganisms (Bandyopadhyay et al., 2009). The rating scale for Organic Carbon suitability used in this study was adopted from the one published by Sys et al. (1991)

2.3.2.5. Land Use and Land Cover

The Land Use and Land Cover (LULC) data is used to evaluate features of an area, including vegetation, urban structures and water bodies. The LULC maps of Bavaria were created using Sentinel-2 imagery. Four images obtained from ArcGIS Living Atlas were processed into a single mosaic with the Mosaic to New Raster tool on ArcGIS. The LULC was classified into five distinct classes, as outlined by Radočaj et al. (2020). Areas in the Vegetation class were identified as optimum locations for soybean cultivation.

2.3.2.6. Climate

According to FAO (2007) soybean requires a minimum temperature of 15 °C and a precipitation range of 450 to 700mm for optimum production. The temperature data obtained from DWD (2024) was given in the form of 1/10 °C to allow for more precise reporting. For this study, the raster calculator was used to convert the temperature values into their decimal form, as shown in the expression below

("temp_extract" / 10)

2.4. Analytical Hierarchy Process (AHP)

The Analytical Hierarchy Process (AHP) method is a popular and widely used method for classifying variables arranged in a hierarchical fashion (Sadiq et al., 2023). It is described as one of the best techniques for determining weight in a multicriteria analysis (Musakwa, 2018). In this study, views from previous research on soybean cultivation were utilised to compare the criteria and assign weight values, which were subsequently processed using the AHP model. The relative ranking for every variable was done using a scale presented by Saaty (1986), ranging from one to nine as shown in Table 3. The weights of the pairwise comparison matrix priorities were then calculated.

Table 3. Scale used for pairwise comparison in AHP.

Intensity of importance	Explanation of score
1	Two attributes having equal importance
3	Weak importance of one over the other
5	Essential or strong importance of one judgment over another
7	Very strong importance of one judgment over another
9	Clear and Extreme preference of one attribute over another
2, 4, 6 and 8	Intermediate values between the two adjacent judgments

Source: (Saaty, 1986, 2004) and (Sadiq et al., 2023)

To assess the consistency of the pairwise comparisons done, the consistency ratio (CR) was computed using Equation 5. A CR score less than 0.1 indicates a desired level of consistency (Sadiq et al., 2023). Equation 4 was used to calculate the consistency index (CI), which is an important consideration when assessing crop land suitability.

$$CI = \frac{\lambda_{max} - n}{n - 1} \quad (4)$$

Here, λ represents the pairwise comparison matrix's greatest eigenvalue, while n indicates the number of classes (Saaty, 1986). The CR was then calculated using Equation 5, following the proposition by Saaty (1986)

$$CR = \frac{CI}{RI} \quad (5)$$

where RI is the ratio index over the average value of CI for random matrices, which was predefined using the Saaty scale (Saaty, 2004).

3. Result

3.1. Reclassification assessments

The criteria used in this land suitability analysis were adjusted and classified into different suitability categories, which are highly suitable (S1), moderately suitable (S2), marginally suitable (S3), currently not suitable (N1), and permanently not suitable (N2). The maps for the reclassification are shown in Figure 3, and Table 4 shows the result for the criteria reclassification. A similar colour scale was used for all the reclassified criteria to aid visualisation.

The results from the soil pH reclassification show that 3.23% of the area falls within the S1 class, 13.99% falls within the S2 class, and 33.57% falls within the S3 class. The N1 class covers 27.29% of the total area, while 21.92% of the study area represents the N2 class.

The organic carbon (OC) reclassification revealed that 0.97% of the study area is within the S1 class, 2.31% falls within the S2 class, and 5.24% falls within the S3 class. 48.79% of the study area represents the N1 class, while 42.68% falls within the N2 class.

Examining the slope reclassification, 23.51% of the total area represents the S1 class, 16.73% of the study area represents the S2 class, and 12.78% of the study area represents the S3 class. The N1 class covers 9.72% of the total area, while 37.19% of the study area falls in the N2 class.

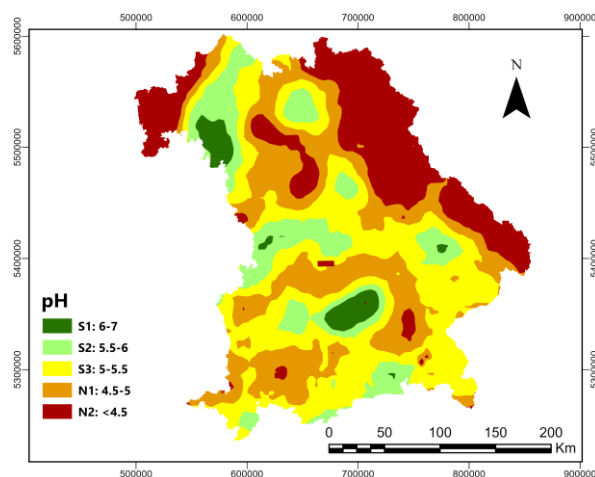
The results of the drainage reclassification show that 61.53% of the total area falls within the S1 class, 2.45% of the total area represents the S2 class, and 0.32% of the total area falls within the S3 class. 0.03% of the study area represents the N1 class, while 35.85% of the total area falls within the N2 class.

For the temperature reclassification, 61.49% of the study area covers the S1 class, 37.70% of the study area covers the S2 class, and 0.45% covers the S3 class. The result also reveals that 0.11% of the study area falls within the N1 class, while 0.01% of the total area represents the N2 class.

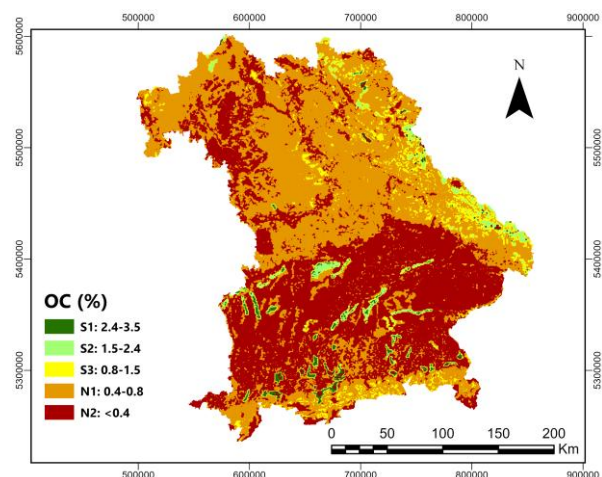
The result of the precipitation reclassification reveals that 57.84% of the study area falls within the S1 class, 27.92% of the study area falls within the S2 class, and 0% of the study area represents the S3 class. 10.30% of the total area represents the N1 class, while 3.71% of the study area represents the N2 class.

The reclassification of LULC shows that 45.78% of the total area represents the S1 class, 43.20% of the total area represents the S2 class, and 0.11% of the total area represents the S3 class. 0.02% of the study area falls within the N1 class, while 10.99% of the study area falls within the N2 class.

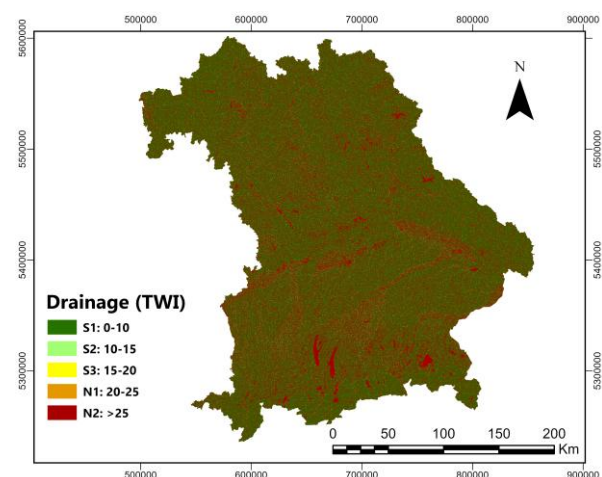
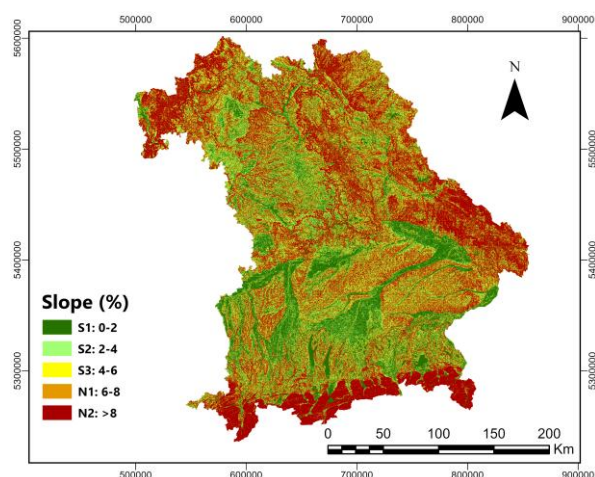
The NDVI reclassified map was not used for the overlay analysis but was used to carry out indirect validation of the final land suitability map.



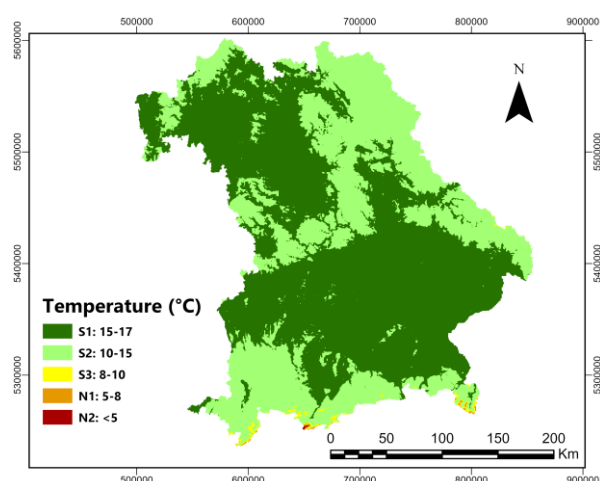
(a) Soil pH reclassification map



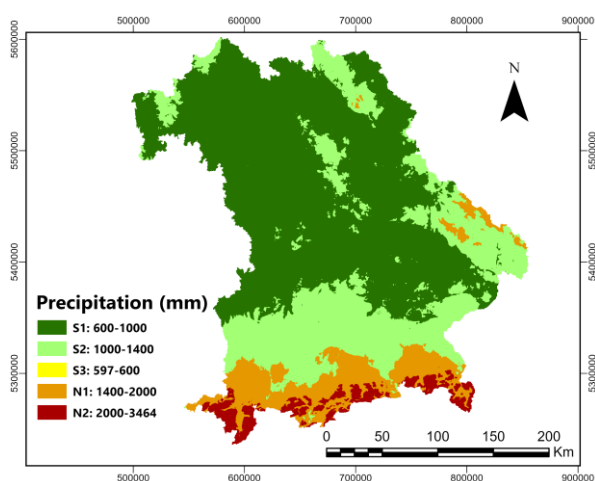
(b) Soil Organic Carbon reclassification map



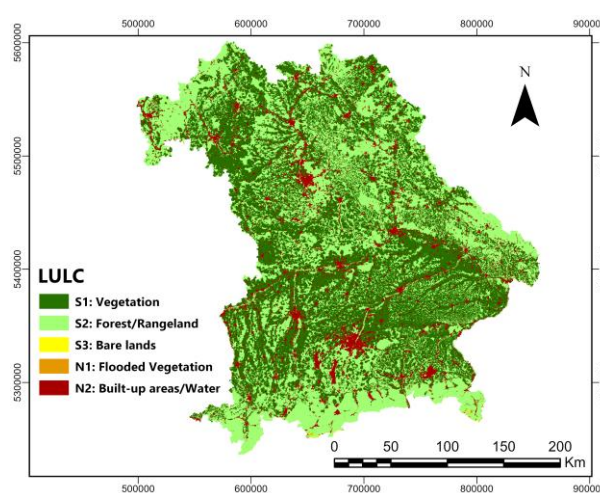
(c) Slope reclassification map



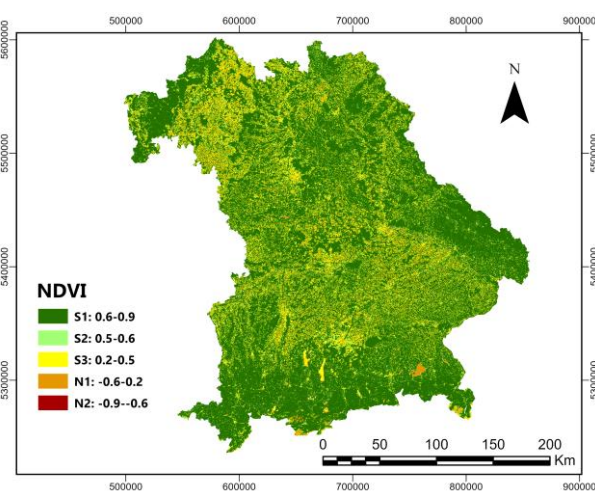
(d) Drainage reclassification map



(e) Temperature reclassification map



(f) Precipitation reclassification map



(g) LULC reclassification map

(h) NDVI reclassification map

Figure 3. a-h: The suitability map of the reclassified criteria for soybean cultivation in Bavaria.

Table 4. Reclassification of criteria data for Soybean suitability analysis.

Criteria	Suitability Class	Area in Hectares	Area in Percentage (%)
Soil pH	S1	227877.81	3.23
	S2	986857.75	13.99
	S3	2368130.75	33.57
	N1	1924751.75	27.29
	N2	1546355.69	21.92
OC	S1	68772.38	0.97
	S2	163292.56	2.31
	S3	369437.56	5.24
	N1	3441433.63	48.79
	N2	3010725.06	42.68
Slope	S1	1658393.31	23.51
	S2	1179981.38	16.73
	S3	901292.25	12.78

Criteria	Suitability Class	Area in Hectares	Area in Percentage (%)
	N1	685326.50	9.72
Slope	N2	2623260.88	37.19
Drainage	S1	4327820.63	61.35
	S2	172737.31	2.45
	S3	22484.25	0.32
	N1	2369.31	0.03
	N2	2528562.25	35.85
Temperature	S1	4337794.94	61.49
	S2	2659379	37.70
	S3	31829.75	0.45
	N1	7504.25	0.11
	N2	1021.19	0.01
Precipitation	S1	4080207.56	57.84
	S2	1969674.25	27.92
	S3	14.25	0
	N1	726249.69	10.3
	N2	261625.81	3.71
LULC	S1	3229513.31	45.78
	S2	3040055.56	43.10
	S3	7704.38	0.11
	N1	1267.19	0.02
	N2	775425.56	10.99

3.2. AHP Weights

After reviewing relevant literature on the cultivation of soybeans, seven criteria were chosen for this land suitability analysis. A pairwise comparison matrix was used to compare the criteria and assign weight values. Each criterion layer in the weighted overlay analysis was then given a priority based on the weight values derived from previous research works.

From the AHP weight results shown in Table 5 and Figure 4, it was observed that “LULC” has the highest cumulative weight index (CWI) value of 0.264, followed very closely by “Drainage” with a CWI value of 0.259. This implies that the Land Use and Drainage have the most significant influence on the suitability analysis. The variables “Slope” and “Precipitation” are the next highest contributors to the suitability analysis, with CWI values of 0.162 and 0.129, respectively. The variables “pH” and “Organic Carbon” have lower CWI values of 0.086 and 0.063, respectively. The variable “Temperature” has a CWI value of 0.037, indicating that it has relatively less importance compared to the other variables. The AHP analysis also revealed a cumulative index (CI) of 0.037 and a consistency ratio of 0.028 (2.8%). This confirms that the pairwise comparison done was highly consistent and logically valid (Sadiq et al., 2023).

Table 5. Results of AHP for evaluating the land suitability of soybean cultivation.

PAIRWISE COMPARISON MATRIX								
	pH	OC	Slope	Drainage	LULC	Temperature	Precipitation	CWI
pH	1	2	0.5	0.25	0.33	3	0.5	0.086
OC	0.5	1	0.33	0.25	0.25	3	0.33	0.063
Slope	2	3	1	0.5	0.5	4	2	0.162
Drainage	4	4	2	1	1	5	2	0.259
LULC	3	4	2	1	1	5	3	0.264
Temperature	0.33	0.33	0.25	0.2	0.2	1	0.25	0.037
Precipitation	2	3	0.5	0.5	0.33	4	1	0.129
CONSISTENCY RESULT								
Consistency Index (CI)								0.037
Ratio Index (RI)								1.32
Consistency Ratio(CR) = CI/RI								0.028

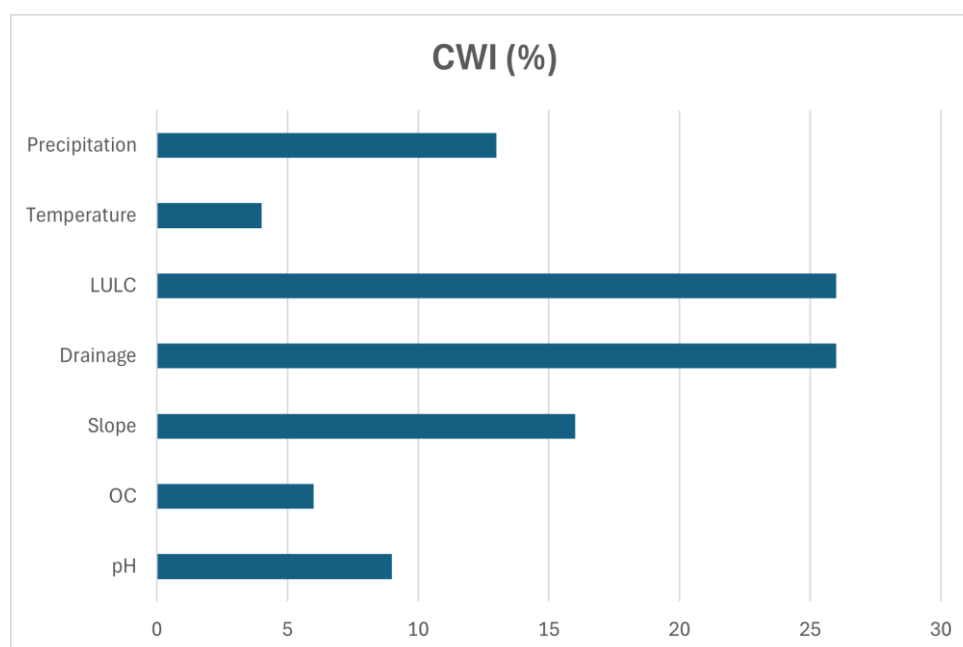


Figure 4. Contribution of each criterion used for AHP in percentage

3.3. Overall Land Suitability Assessment

The evaluation of land suitability for the cultivation of soybeans in Bavaria involved important factors, including soil pH, soil organic carbon, slope, drainage, land use/land cover, temperature and precipitation. The final land suitability map, as shown in Figure 5, was created by overlaying seven layers corresponding to the criteria listed in Table 5. Figure 5 shows the overall land suitability map specifically produced for soybean farming in the Free State of Bavaria, Germany.

An overview of the suitability classes of the study area measured in hectares (ha) and percentages (%) is shown in Table 6. The suitability classes identified are “Highly Suitable”, “Moderately Suitable”, “Marginally Suitable”, “Currently Not Suitable”, and “Permanently Not Suitable”. The total area covered in this suitability assessment is approximately 7,033,045.94 ha.

Table 6 shows that the Highly Suitable class covers an area of 995725.69 ha, which accounts for approximately 14.16% of the total study area. The Moderately Suitable class covers a much larger area of 3860134.69 ha, which corresponds to approximately 54.89% of the total area. This indicates that a significant portion of Bavaria falls into this category. The Marginally Suitable class covers an area of 1956074.31 ha, accounting for approximately 27.81% of the total area. The Currently Not Suitable and Permanently Not Suitable classes cover an area of 218863 ha and 2248.25 ha, respectively, both accounting for 3.11% and 0.03% respectively, of the total study area.

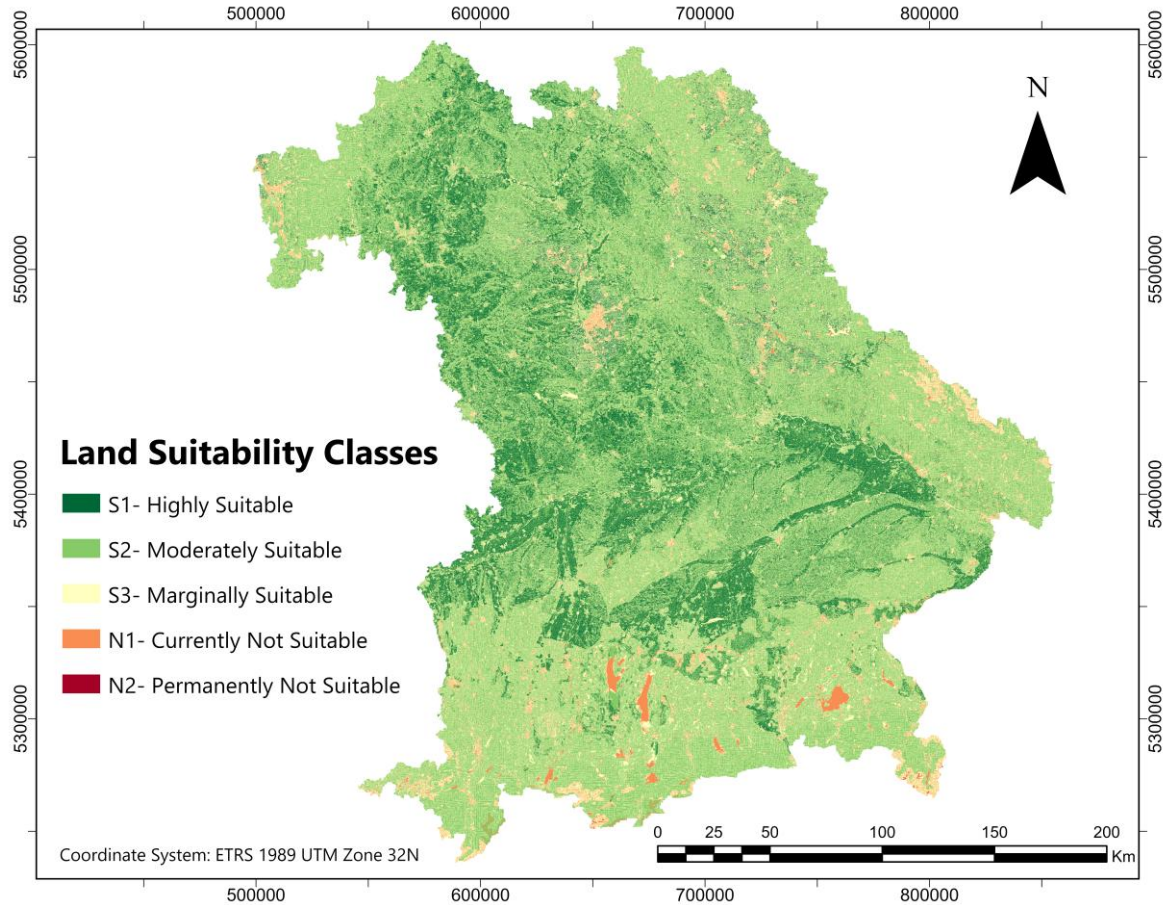


Figure 5. Land suitability map for soybean cultivation in Bavaria, Germany.

Table 6. Land suitability distribution for soybean cultivation in Bavaria, Germany

Class Code	Suitability class	Area (ha)	Area (%)
S1	Highly Suitable	995725.69	14.16
S2	Moderately Suitable	3860134.69	54.89
S3	Marginally Suitable	1956074.31	27.81
N1	Currently Not Suitable	218863.00	3.11
N2	Permanently Not Suitable	2248.25	0.03

3.4. Validation of the Suitability Map

The final land suitability map was validated by an indirect method, as proposed by Radočaj et al. (2020) using the Normalized Difference Vegetation Index (NDVI) of Sentinel-2 imagery with minimal cloud cover, captured during the soybean seed filling growth stage (R6), which is usually in August. This was done by comparing the suitability classes of the final suitability map with areas showing healthy vegetation growth as indicated by the NDVI values as shown in the NDVI Reclassified map in Figure 3h.

The indirect validation primarily targets the Highly Suitable class of the final suitability map. The NDVI map in Figure 3h shows that, when the satellite imagery was captured, during the peak season for soybean growth, about 70.9% of the study area fell within the S1 class, which are areas with very high NDVI values. This indicates areas that had high photosynthetic activity, dense canopy and healthy growth and could subsequently be categorised as potentially optimal areas for cultivating soybean (Radočaj et al., 2020).

The Tabulate Area geoprocessing tool was used to determine the area of spatial overlap between the final suitability zones and the five NDVI reclassified classes. The final validation result obtained from ArcGIS is shown in Table 7. The total highly suitable region of the final suitability map that overlapped with the NDVI map is 608206.31 ha, which corresponds to approximately 61.08% of the areas initially identified as highly suitable areas.

Table 7. Validation result based on comparison with peak NDVI values

Suitability Class	Suitability Class Area	Areas that overlap	% of overlap
Highly Suitable	995725.69	608206.31	61.08
Moderately Suitable	3860134.69	256201.06	6.64
Marginally Suitable	1956074.31	292496.81	14.95
Currently Not Suitable	218863.00	24952.89	11.40
Permanently Not Suitable	2248.25	4.69	0.21

4. Discussion

Bavaria generally has some level of variation in the different factors and criteria considered in this study. The study area's pH levels ranged from highly acidic to neutral. The quick loss of basic cations from the surface soil caused by erosion may be the cause of this pH fluctuation (Sadiq et al., 2023). Therefore, adding acidic fertilisers to regions of Bavaria with acidic soils may reduce the amount of nutrients available for plant uptake. For the soil's organic carbon, Bavaria has very limited areas with substantial organic matter in its soil. The majority of soils in Bavaria have less than 0.8% organic carbon content. Hence, there will be a great need to include fertilizer during the cultivation of soybeans in Bavaria.

Concerning slope classification, approximately 24% of the land in Bavaria is suitable for the cultivation of soybeans. However, a much greater portion of Bavaria is not exactly suitable for the cultivation of soybeans. Also, in terms of soil drainage, soybeans will thrive in approximately 60% of areas in Bavaria due to the good soil drainage conditions in the region.

The LULC data indicates that about 45% of the total area is suitable for soybean cultivation. This stands as a major advantage for the state of Bavaria. One of the main causes of land loss that has been identified is urbanisation (Radočaj et al., 2020). However, this is not the case for Bavaria, as it has just about 11% of its area categorized as built-up areas. Bavaria also has a climate that supports the cultivation of soybeans. The rainfall and precipitation data prove that the climate of Bavaria matches the requirements specified by (FAO).

Overall, Table 6 shows the distribution parameters across the different suitability classes. Based on Figure 5, the highly suitable zone is clustered in the central and south-central region of the study. There is also a visible patch of highly suitable zones in the northern extent of the study area. The study area is covered mainly by moderately suitable and marginally suitable areas. This indicates that while most of Bavaria can potentially support the cultivation of soybeans, many areas have one or two limiting factors.

More so, the validation result shows that about 61% of the areas classified as highly suitable are potentially accurate. This indicates that the multicriteria analysis carried out using factors like soil pH,

slope, and climate has truly identified areas that are considerably capable of supporting healthy and successful soybean growth. Hence, for the farmers in this area to produce soybeans with maximum yield, they need to reduce the identified limitations of the study area.

5. Conclusion

This study assessed the land suitability for soybean cultivation in Bavaria, Germany, to support sustainable agricultural planning. This research used the Analytical Hierarchy Process (AHP) to derive expert-based criteria weights, integrating them with Geographic Information Systems (GIS) techniques for spatial evaluation. This analysis considered six key suitability parameters, with the results categorized using FAO guidelines. The assessment indicated drainage and LULC as the most influential factors, followed by slope, pH, and Organic Carbon (OC). The final suitability map revealed that the Highly Suitable class covered 14.16% of the region, the Moderately and Marginally Suitable classes covered 54.89% and 27.81% of the region. Also, the Unsuitable Areas accounted for 3.14% of the study area. Indirect validation using NDVI overlap confirmed the predictive power of the suitability analysis, showing a strong positive correlation between highly suitable zones and areas of healthy vegetation growth.

The successful integration of AHP and GIS provides local stakeholders, including government and farmers, with a reliable tool for informed land-use planning and targeted management to optimize future soybean productivity in Bavaria.

References

- Aragónés-Beltrán, P., Chaparro-González, F., Pastor-Ferrando, J. P., & Pla-Rubio, A. (2014). An AHP (Analytic Hierarchy Process)/ANP (Analytic Network Process)-based multi-criteria decision approach for the selection of solar-thermal power plant investment projects. *Energy*, 66, 222–238. <https://doi.org/10.1016/j.energy.2013.12.016>
- Bandyopadhyay, S., Jaiswal, R. K., Hegde, V. S., & Jayaraman, V. (2009). Assessment of land suitability potentials for agriculture using a remote sensing and GIS based approach. *International Journal of Remote Sensing*, 30(4), 879–895. <https://doi.org/10.1080/01431160802395235>
- Bedawi, G., Balasundram, S. K., Shariff, A. R. M., & Abdullah, A. F. bin. (2016). Agriculture land suitability analysis evaluation based multi criteria and GIS approach. *8th IGRSM International Conference and Exhibition on Remote Sensing & GIS (IGRSM 2016)*, July. <https://doi.org/10.1088/1755-1315/37/1/012044>
- Bogenreuther, J., Bogner, C., Siebert, S., & Koellner, T. (2025). Heat, drought, and compound events: Thresholds and impacts on crop yield variability. *Agricultural and Forest Meteorology*, 375(May), 110836. <https://doi.org/10.1016/j.agrformet.2025.110836>
- Böttcher, T., & Zimmer, Y. (2021). Expanding Crop Rotations Profitably with Soybean Cultivation. *Thünen Working Paper 169*.
- Coulter, J., & Moncada, K. (2008). Soybean Production. In *RISK MANAGEMENT GUIDE FOR ORGANIC PRODUCERS* (pp. 1–18).
- de Brogniez, D., Ballabio, C., Stevens, A., Jones, R. J. A., Montanarella, L., & van Wesemael, B. (2015). A map of the topsoil organic carbon content of Europe generated by a generalized additive model. *European Journal of Soil Science*, 66(1), 121–134. <https://doi.org/10.1111/ejss.12193>
- de la Casa, A., Ovando, G., Bressanini, L., Martínez, J., Díaz, G., & Miranda, C. (2018). Soybean crop coverage estimation from NDVI images with different spatial resolution to evaluate yield

- variability in a plot. *ISPRS Journal of Photogrammetry and Remote Sensing*, 146(October), 531–547. <https://doi.org/10.1016/j.isprsjprs.2018.10.018>
- DWD 2023. Deutscher Wetterdiens. (n.d.). *CDC Climate Data Cente*. Retrieved October 28, 2025, from <https://cdc.dwd.de/>
- DWD Climate Data Center (CDC). (2024a). *Annual sum of monthly precipitation grids over Germany, version v1.0*. <http://www.dwd.de/>
- DWD Climate Data Center (CDC). (2024b). *Annual sum of monthly temperature grids over Germany*. <http://www.dwd.de/>
- EEA 2023. European Environment Agency. (n.d.). *Euroeapn Digital Elevation Model (EUEDEM), version 1.0*. Retrieved October 28, 2025, from European Digital Elevation Model (EUEDEM), version 1.0.
- Environmental Systems Research Institute (ESRI). (n.d.). *Pixel size of image and raster data*. ArcGIS Pro 3.5 Help. Retrieved November 6, 2025, from <https://pro.arcgis.com/en/pro-app/latest/help/data/imagery/pixel-size-of-image-and-raster-data-pro-.htm>
- European Soil Data Centre (ESDAC). (2010). *European Soil Database & soil properties*. <https://esdac.jrc.ec.europa.eu/resource-type/european-soil-database-soil-properties>
- Fogelberg, F., & Recknagel, J. (2017). Developing Soy production in Central and Northern Europe. *Legumes in Cropping Systems*, 109–124. <https://doi.org/10.1079/9781780644981.0109>
- Food and Agriculture Organization of the United Nations (FAO). (2007). Land Evaluation - Towards a revised framework. In *FAO* (p. 124).
- GADM. (2025). *Database of Global Administrative Areas*. <https://gadm.org/data.html>
- Gebhardt, S., Haensel, M., Schulp, C. J. E., & Kaim, A. (2022). Ecologically and biophysically optimal allocation of expanded soy production in Bavaria, Germany. *Frontiers in Sustainable Food Systems*, 6. <https://doi.org/10.3389/fsufs.2022.916003>
- Geng, S., Shi, P., Zong, N., & Zhu, W. (2019). Agricultural Land Suitability of Production Space in the Taihang Mountains, China. *Chinese Geographical Science*, 29(6), 1024–1038. <https://doi.org/10.1007/s11769-019-1075-6>
- Ishizaka, A., & Labib, A. (2009). Analytic Hierarchy Process and Expert Choice: Benefits and limitations. *OR Insight*, 22(4), 201–220. <https://doi.org/10.1057/ori.2009.10>
- Jenson, S. K., & Domingue, J. O. (1988). Extracting topographic structure from digital elevation data for geographic information system analysis. *Photogrammetric Engineering and Remote Sensing*, 54(11), 1593–1600.
- Khormali, F., Ayoubi, S., Foomani, F. K., Fatemi, A., & Hemmati, K. (2007). Tea yield and soil properties as affected by slope position and aspect in Lahijan area, Iran. *International Journal of Plant Production*, 1(1), 98–111. www.ijpp.info
- Kumar, S., Patel, N. R., Sarkar, A., & Dadhwal, V. K. (2013). Geospatial Approach in Assessing Agro-Climatic Suitability of Soybean in Rainfed Agro-Ecosystem. *Journal of the Indian Society of Remote Sensing*, 41(3), 609–618. <https://doi.org/10.1007/s12524-012-0249-9>
- Liu, Z. jun, Zhou, W., Shen, J. bo, Li, S. tian, Liang, G. qing, Wang, X. bin, Sun, J. wen, & Ai, C. (2014). Soil Quality Assessment of Acid Sulfate Paddy Soils with Different Productivities in

- Guangdong Province, China. *Journal of Integrative Agriculture*, 13(1), 177–186.
[https://doi.org/10.1016/S2095-3119\(13\)60594-8](https://doi.org/10.1016/S2095-3119(13)60594-8)
- Malczewski, J. (2004). GIS-based land-use suitability analysis: A critical overview. *Progress in Planning*, 62(1), 3–65. <https://doi.org/10.1016/j.progress.2003.09.002>
- Michels, M., & Wever, H. (2025). German farmers' perceptions of soybean cultivation -A Q-methodology analysis. *Research Square*. <https://doi.org/10.21203/rs.3.rs-7367959/v1>
- Montgomery, B., Dragičević, S., Dujmović, J., & Schmidt, M. (2016). A GIS-based Logic Scoring of Preference method for evaluation of land capability and suitability for agriculture. *Computers and Electronics in Agriculture*, 124, 340–353. <https://doi.org/10.1016/j.compag.2016.04.013>
- Munene, P., Chabala, L. M., & Mweetwa, A. M. (2017). Land Suitability Assessment for Soybean (*Glycine max* (L.) Merr.) Production in Kabwe District, Central Zambia. *Journal of Agricultural Science*, 9(3), 74. <https://doi.org/10.5539/jas.v9n3p74>
- Musakwa, W. (2018). Identifying land suitable for agricultural land reform using GIS-MCDA in South Africa. *Environment, Development and Sustainability*, 20(5), 2281–2299.
<https://doi.org/10.1007/s10668-017-9989-6>
- Perveen, S., Arsalan, M. H., & Siddiqui, M. F. (2013). *Gis-Based Multi-Criteria Model for Cotton Crop Land Suitability : a Perspective From Sindh Province of Pakistan*. 3(May 2014), 31–37.
- Radočaj, D., Jurišić, M., Gašparović, M., & Plaščak, I. (2020). Optimal soybean (*Glycine max* L.) land suitability using gis-based multicriteria analysis and sentinel-2 multitemporal images. *Remote Sensing*, 12(9). <https://doi.org/10.3390/RS12091463>
- Radocaj, D., Jurišić, M., Zebec, V., & Plaščak, I. (2020). Delineation of soil texture suitability zones for soybean cultivation: A case study in continental Croatia. *Agronomy*, 10(6).
<https://doi.org/10.3390/agronomy10060823>
- Rosa, D. la, & C.A., van D. (2009). Qualitative and quantitative physical land evaluation: An operational approach. *UNESCO-EOLSS Encyclopedia, Oxford, UK, July*, 195.
- Saaty, T. L. (1986). *Axiomatization of the Analytic Hierarchy Process* (pp. 91–108).
https://doi.org/10.1007/978-3-642-46536-9_4
- Saaty, T. L. (2004). Decision making — the Analytic Hierarchy and Network Processes (AHP/ANP). *Journal of Systems Science and Systems Engineering*, 13(1), 1–35. <https://doi.org/10.1007/s11518-006-0151-5>
- Sadiq, F. K., Ya'u, S. L., Aliyu, J., & Maniyunda, L. M. (2023). Evaluation of land suitability for soybean production using GIS-based multi-criteria approach in Kudan Local Government area of Kaduna State Nigeria. *Environmental and Sustainability Indicators*, 20(July), 100297.
<https://doi.org/10.1016/j.indic.2023.100297>
- Schmidt, F., & Persson, A. (2003). Comparison of DEM data capture and topographic wetness indices. *Precision Agriculture*, 4(2), 179–192. <https://doi.org/10.1023/A:1024509322709>
- Shekar, S., Xiong, H., & Zhou, X. (2017). *Exploratory Data Analysis BT - Encyclopedia of GIS*.
https://doi.org/10.1007/978-3-319-17885-1_100394
- Sys, C., Van Ranst, E., & Debaveye, J. (1991). Land Evaluation, Part I: Principles in Land Evaluation and Crop Production Calculations. In *Agricultural Publications-N0 7, General Administration for Development Cooperation, Brussels, Belgium* (p. 280).

Weinzierl, M. (2019). *Study on the market for soy beans in Germany*.

Wiesmeier, M., Burmeister, J., Garcia-Franco, N., & Sümmerer, M. (2025). Long-term changes (1986–2016) in soil organic carbon and nitrogen in agricultural soils of Bavaria. *Agriculture, Ecosystems and Environment*, 394(June 2024), 109874. <https://doi.org/10.1016/j.agee.2025.109874>

Wilson, J. P., & Gallant, J. C. (2000). Digital Elevation Models and Representation of Terrain Shape. In *Terrain Analysis: Principles and Applications* (pp. 29–50).

Zepp, S., Heiden, U., Bachmann, M., Möller, M., Wiesmeier, M., & van Wesemael, B. (2023). Optimized bare soil compositing for soil organic carbon prediction of topsoil croplands in Bavaria using Landsat. *ISPRS Journal of Photogrammetry and Remote Sensing*, 202(November 2022), 287–302. <https://doi.org/10.1016/j.isprsjprs.2023.06.003>